Characterization of Exponentially Modified Gaussian Peaks in Chromatography

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The exponentially modified Gaussian peak has attracted attention recently in computer deconvolution of chromatographic peaks. This peak shape model has some theoretical justification. In this paper, we investigate the behavior of the skew and excess and of the second derivative of that shape in order to examine their utility in the analysis of strongly overlapped chromatographic peaks. The study showed that, indeed, moment as well as slope analysis can be beneficial in characterizing double peaks. In addition, moment analysis of a single peak can indicate the magnitude of extra-column effects. If the time constant of these effects is known, slope analysis might be preferred because of its simplicity, to moment analysis.

THE PROBLEM of strongly overlapped chromatographic peaks can be dealt with by employing statistical moments or slope analysis (*1*, *2*). In the former (*1*), deviations in the peak shape, as obtained from the skew and excess, between a single peak and a double peak in the form of a single band are used as the indicator of a strongly overlapped system. In the latter (*2*), deviation in the second derivative behavior, when manipulated appropriately, ascertains the existence of double peaks. In discussing these two methods, the mathematical models which were described were developed mainly because of their computational ease. Although the models were realistic, they were chosen arbitrarily (except for the "kinetic tailing" model). A better theoretical model for the chromatographic peak shape is thus needed.

Exponentially modified Gaussian peaks, that is to say, a Gaussian convoluted with an exponential decay function, seem to fill that need. More than a decade ago, Schmauch (*3*) as well as Johnson and Stross (*4*) recognized that instrumental contribution, such as detector dead volume, will modify exponentially the chromatographic peak. Esser (*5*), in investigating spectrophotometric detectors, reached the same conclusion. Sternberg, in his comprehensive review on extra-column effects (*6*), shows that various dead volume contributions alter the peak exponentially. He indicates that each particular extra-column effect has a certain time constant τ associated with it. The magnitude of this time constant determines the extent to which the peak is distorted. More recently, Gladney *et al.* (*7*) used the exponentially modified Gaussian in numerical deconvolution techniques. They have shown that experimental peaks can be described reasonably well by this model. The same shape was used by Littlewood and coworkers (*8*) in their deconvolution method. In addition they have briefly investigated the effect of mixing chambers and have concluded that these affect the skewness of the peak. McWilliam and Bolton (*9*) derived the exponentially convoluted Gaussian in their discussion on instrumental peak distortion, and its effect on the resolution. More recently (*10*) they used this peak model in further discussion on the area recovery of two partially overlapped peaks.

With this interest in exponentially modified Gaussians, it seems useful to investigate the theoretical behavior of two strongly overlapped (*i.e.*, resolution >0.5) such peaks. This is particularly so since this peak model has some theoretical and experimental (*7*, *8*) justification.

The exponentially modified Gaussian peak is defined by the following convolute integral.

$$f(t) = \frac{A}{\tau\sigma\sqrt{2\pi}} \int_0^{\infty} \exp\left[-\frac{(t - t_R - t')^2}{2\sigma^2}\right] \exp\left[\frac{-t'}{\tau}\right] dt' \quad (1)$$

Or, alternatively

$$f(t) = \frac{A\sigma}{\tau\sqrt{2}} \exp\left[\left(\frac{\sigma}{\tau}\right)^2 \frac{1}{2} - \frac{(t - t_R)}{\tau}\right] \int_{-\infty}^{Z} \exp[-x^2]dx \quad (2)$$

where $Z = [(t - t_R)/\sigma - \sigma/\tau]1/\sqrt{2}$ in Equation 2, A is the peak amplitude, σ is the variance of the Gaussian, t_R is the center of gravity of the Gaussian, τ is the time constant of the exponential modifier (which can be attributed, among others, to extra column contribution) and t' and x are dummy variables of integration. Some of the properties of these expressions were discussed by Gladney *et al.* (*7*) and by McWilliam and Bolton (*9*). It should be noted that the area of this convolution expression is equal to that of a pure Gaussian and that the maximum of the peak always falls on the Gaussian which is being modified (*9*). Moreover, the asymmetry of the peak depends on the ratio τ/σ. Representative examples of peaks with various τ/σ values can be found elsewhere (*7–9*). Before proceeding to investigate, theoretically, the behavior of the skew and excess, as well as that of the second derivative, of a single and double convoluted peaks, some comments should be made concerning some practical aspects of these analyses. The experimental methodology of obtaining the moments or the derivatives was already discussed by us (*1*, *2*). The experimental accuracies reported are well within the reach of many laboratories, especially when calibration curves are used as standards.

MOMENT ANALYSIS

The moments, M_n, of Equation 1 can be found by employing the Laplace (or the Fourier) transformation. Use is being

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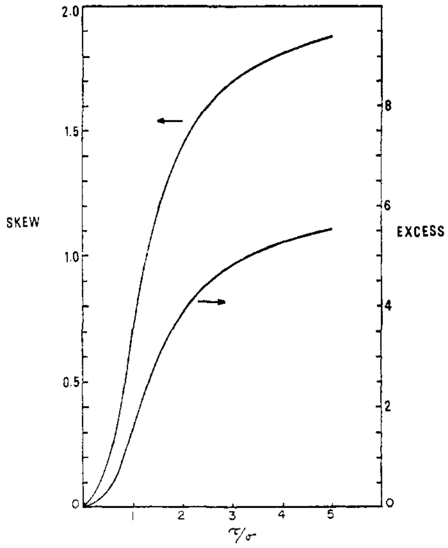

Figure 1. Behavior of the skew and the excess as a function of τ/σ

made here of the following property of the Laplace transform of a convolution $f(t)$:

$$\bar{f}(s) = \int_0^\infty e^{-st}\int_0^\infty g(t')h(t - t')dt'dt = \bar{g}(s)\cdot\bar{h}(s) \quad (3)$$

The bar indicates a Laplace transform and s is the Laplace variable. The moments of $f(t)$ are found from the relation

$$M_n = \lim_{s\to 0}\left(\frac{d^n\bar{f}(s)}{ds^n}\right)\left(\frac{1}{\bar{f}(s)}\right) \quad (4)$$

Alternatively, the moments can be obtained by integrating the convolution expression (Equation 1) according to the definition of the moments:

$$M_n = \frac{1}{M_0}\int_{-\infty}^{\infty} t^n \int_0^\infty \frac{A}{\sigma\tau\sqrt{2\pi}} \times \exp\left[-\frac{(t - t_R - t')^2}{2\sigma^2}\right]\exp\left(\frac{-t'}{\tau}\right)dt'dt \quad (5)$$

when M_0 is the zeroth moment or the peak area.

This method of obtaining the moments is simpler than the Laplace method. Both integrands in the convolution equation are well behaved throughout the integration region so that the order of integration can be interchanged. We thus integrate Equation 5 first with respect to t and then with respect to t'.

The first moment of the convoluted integral is

$$M_1 = t_R + \tau \quad (6)$$

Moments higher than the first are central moments (*i.e.*, taken around M_1). The second, third, and fourth central moments are:

$$M_2 = \sigma^2 + \tau^2 \quad (7)$$

$$M_3 = 2\tau^3 \quad (8)$$

$$M_4 = 3\sigma^4 + 6\sigma^2\tau^2 + 9\tau^4 \quad (9)$$

The skew and excess of a single convoluted peak are:

$$\text{Skew} = \frac{M_3}{M_2^{3/2}} = \frac{2\left(\frac{\tau}{\sigma}\right)^3}{\left[1 + \left(\frac{\tau}{\sigma}\right)^2\right]^{3/2}} \quad (10)$$

$$\text{Excess} = \frac{M_4}{M_2^2} - 3 = \frac{3(1 + 2\tau^2/\sigma^2 + 3\tau^4/\sigma^4)}{\left[1 + \left(\frac{\tau}{\sigma}\right)^2\right]^2} - 3 \quad (11)$$

When τ/σ is zero, the skew and the excess are zero since the peak is a Gaussian one. In the other limit as τ/σ goes to infinity, the peak is pure exponential decay and the skew and excess are:

$$\lim_{\frac{\tau}{\sigma}\to\infty} \text{Skew} \to 2.0 \quad (12)$$

$$\lim_{\frac{\tau}{\sigma}\to\infty} \text{Excess} \to 6.0 \quad (13)$$

Littlewood (*8*) called the ratio τ/σ the "skew." While this ratio determines the magnitude of the skew, the statistically accepted definition of the skew is given by Equation 10. The behavior of the skew and the excess as a function of τ/σ (up to the value of 5) is shown in Figure 1. The skew changes the fastest (its inflection point) at τ/σ ratio of about 0.81. The maximum rate of change in the excess is around τ/σ = 1.0. It is also interesting to note that if the chromatographic peak produced by the column is a Gaussian one (this can be expected in the "long time" limit of chromatography), the magnitude of the extra-column contribution τ can be obtained directly from the third central moment. Once known, the time variance, σ^2, of the peak due to column processes may be ascertained from the second moment of the exponentially modified peak. In addition, knowledge of the time constant of the extra-column effects can set a criterion as to the narrowest permissible peaks which would be relatively unperturbed by these effects. For example, Figure 1 shows that if a skew of 0.1, or less, can be tolerated, the ratio τ/σ must be less than 0.4. Thus, if τ is 0.1 second, the chromatographic zone produced by the column must have $\sigma \geq 0.25$ sec (or a base width of 1 sec). Narrower peaks will be adversely affected by that time constant. If, on the other hand, τ = 1 sec, the peak should be at least 10 sec wide before being distorted. The ramifications of a "low value" system (minimum extra column contribution), especially in high speed chromatography, are obvious. A similar argument to this one was advanced by McWilliam (*9*).

In the context of this paper, perhaps a more important behavior is that of the skew and excess simultaneously. In Figure 2 the solid line is that generated by the skew and excess as the ratio τ/σ of a single exponentially modified Gaussian peak is varied. In this skew–excess coordinate system, the solid line starts at the origin (where τ/σ = 0) and should increase to the limits shown in Equations 12 and 13. In Figure 2, however, since peaks with τ/σ ratio of about 2.6 and higher are rather distorted, we have extended the solid line only up to the skew and excess values of a peak with that τ/σ ratio. Each point on the solid line belongs to a single exponentially convoluted Gaussian having a particular τ/σ ratio.

Since we are interested in the behavior of strongly overlapped peaks, we have extended the moments expression to a system of two convoluted peaks separated by the distance d. The equation describing the two peaks system is:

$$f(t) = A\int_0^\infty \exp\left[\frac{-(t - t_R - t')^2}{2\sigma_1^2}\right]\exp[-t'/\tau]dt' + B\int_0^\infty \exp\left[\frac{-(t - t_R - t' - d)^2}{2\sigma_2^2}\right]\exp(-t'/\tau)dt' \quad (14)$$

where A and B are the height of each peak and subscript 1 and 2 identify the peak. The zeroth moment (or the composite area) is simply the sum of the areas of the two peaks. The first four moments (moments heigher than the first are central moments) are:

$$M_1 = \frac{t_R(1 + c) + \tau(1 + c) + dc}{(1 + c)} \tag{15}$$

$$M_2 = \frac{\sigma_1^2(1 + c)^2 + \tau^2(1 + c)^2 + d^2c^2}{(1 + c)^3} + c\left[\frac{\sigma_2^2(1 + c)^2 + \tau^2(1 + c)^2 + d^2}{(1 + c)^3}\right] \tag{16}$$

$$M_3 = \frac{2\tau^3(1 + c)^3 - 3d\tau^2c(1 + c)^2 - 3\sigma_1^2dc(1 + c)^2 - c^3d^3}{(1 + c)^4} + \frac{c[2\tau^3(1 + c)^3 + 3d\tau^2(1 + c)^2 + 3\sigma_2^2d(1 + c)^2 + d^3]}{(1 + c)^4} \tag{17}$$

$$M_4 = [3\sigma_1^4(1 + c)^4 + 6\sigma_1^2\tau^2(1 + c)^4 + 9\tau^4(1 + c)^4 + 6\sigma_1^2d^2c^2(1 + c)^2 - 8\tau^3dc(1 + c)^3 + d^4c^4 + 6\tau^2d^2c^2(1 + c)^2]/(1 + c)^5 + c[3\sigma_2^4(1 + c)^4 + 6\sigma_2^2\tau^2(1 + c)^4 + 9\tau^4(1 + c)^4 + d^4 + 8\tau^3d(1 + c)^3 + 6\tau^2d^2(1 + c)^2 + 6\sigma_2^2d^2(1 + c)^2]/(1 + c)^5 \tag{18}$$

c is equal to $B\sigma_2/A\sigma_1$. The skew and excess can be calculated directly from the last three moments.

The behavior of the skew and excess was investigated with the aid of a CDC-6400 computer. At first, we assumed that the two peaks in the composite are of identical width; *i.e.*, identical τ/σ ratio. For strongly overlapped peaks, this assumption is not a bad one and it is made frequently in chromatography. Some of the extra-column contributions, *i.e.*, diffusion chambers, might affect each of the components differently yielding different τ/σ ratios. However, in most cases, closely eluted peaks have similar physical properties, thus making this difference in τ/σ very small. The skew and excess of double peaks having equal τ/σ value were obtained for several τ/σ cases at several peak height ratio. The dashed lines in Figure 2 indicate the behavior of the skew and excess of several composite systems, each composed of two identical peaks (equal τ/σ and equal height), as the separation, d, is varied from zero (indicated by the circles on the solid) to about 1.4 σ units. At d equal to zero, the two peaks in the composite are completely overlapped and the system is equivalent to a single peak. Hence, all the dashed lines begin at the solid line. Figure 2 demonstrates that the skew and the excess of double equal peaks fall to the right of the single peak solid line. In all cases, the skew and excess of the double peaks are sufficiently different from these of a true single peak. However as the ratio τ/σ increases, the skew and excess of a double peak, although different from these of single peak with the same τ/σ ratio, lie close to the single peak line. In fact, as will be shown shortly, if the peaks in the composite are not equal in height, the skew and excess of such a double peak system might fall on, or cross, the solid line. This is particularly true for systems with high τ/σ ratio. As the time constant of the system decreases, or as the peak width increases, this ambiguity in interpreting the solid line is removed. In the limit of a pure Gaussian ($\tau/\sigma = 0$), any system of two peaks of any height ratio and any separation except $d = 0$ can be easily and unmistakably recognized as being a double peak.

To investigate in more details the effect of the peak height ratio we centered our attention on the composite made of two

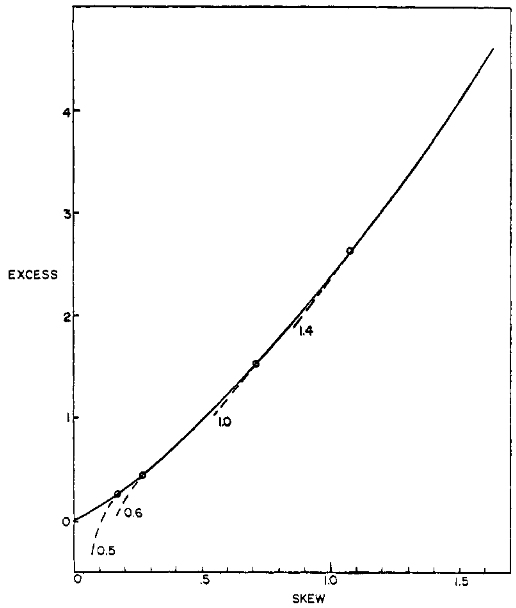

Figure 2. Excess *vs.* skew. Each dashed line corresponds to a system of two identical peaks.

The number next to each dashed line is the τ/σ value of the two peaks in the composite

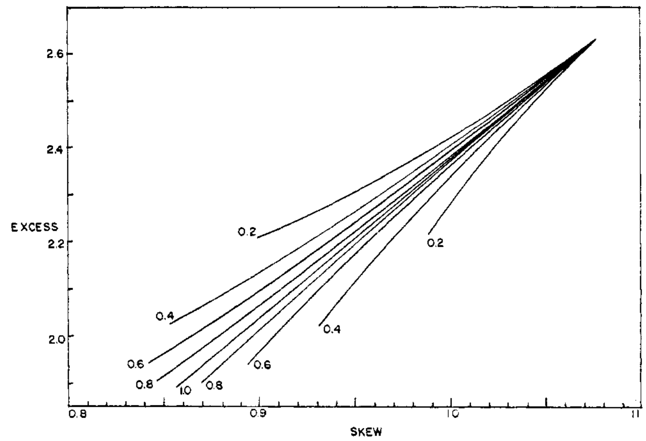

Figure 3. Excess *vs.* skew of a composite in which $\tau/\sigma_1 = \tau/\sigma_2 = 1.4$

The number next to each line indicates the height ratio of the two peaks. The numbers above the line corresponding to $A = B$ (1.0) are A/B values. Numers below that line are B/A values

peaks with $\tau/\sigma = 1.4$. We chose this system for two main reasons: The shape of a single such peak is frequently encountered in actual practice [a drawing of a single peak with $\tau/\sigma = 1.5$ can be found in reference (*8*)]; the ratio τ/σ is high enough so that we can study the above mentioned ambiguity where the double peak system has the same skew and excess as these of a single peak of some different τ/σ. Figure 3 shows the skew and excess of a double peak system for various height ratios of the peaks in the composite. The height ratio of the first to the second peak varies from 0.2 to 5. All the lines (one for each height ratio) emanate from the same point; *i.e.*, that of a single peak with $\tau/\sigma = 1.4$. Each line extends to the point where $d = 1.40$ σ units. This separation corresponds to a resolution of 0.350 in the case of a pure Gaussian. In the system discussed here, however, the resolution is less than 0.350 due to the tailing of the leading peak. Figure 3

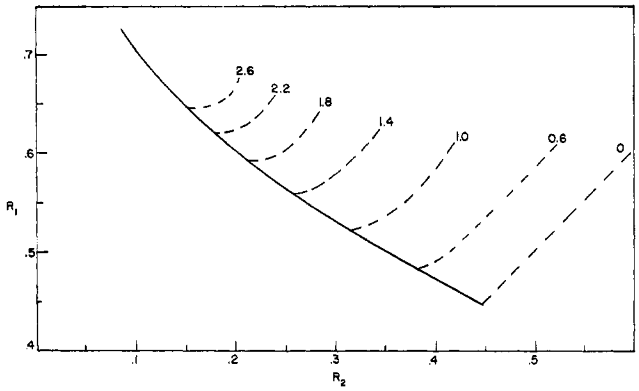

Figure 4. R_1 *vs.* R_2. **Each dashed line corresponds to a system of two identical peaks**

The number next to each dashed line is the τ/σ value of the two peaks in the system

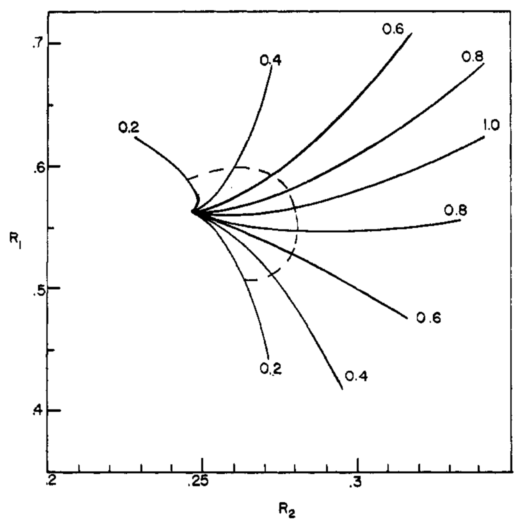

Figure 5. R_1 *vs.* R_2 **of a composite in which** $\tau/\sigma_1 = \tau/\sigma_2 = 1.4$

The number next to each line indicates the height ratio of the two peaks. The number above the line corresponding to $A = B$ (1.0) are B/A values. Numbers below that line are A/B values

indicates that the skew and excess of the double peak, for any height ratio, is sufficiently different from these of a true single peak. The change in the skew and excess with the separation are the most pronounced when the two peaks are about the same height. As either of the peaks in the composite decreases in height, the changes in the skew and excess diminish. This is not surprising, since in the limit where either of the peaks vanishes, the skew and excess of the composite reduce to that of a single peak. Since the peaks in the system are asymmetric, it makes quite a bit of difference whether the leading or the following peak is the smaller of the two. In fact, Figure 3 shows that in the case where the leading peak (indicated by subscript 1 in Equation 14) is smaller, the skew and excess lie above the line corresponding to two equal peaks, while the skew and excess lie below that line when the trailing peak (indicated by subscript 2 in Equation 14) is the smaller one. Figure 3 indicates clearly that the system composed of two peaks with τ/σ ratio of 1.4 is completely defined in this coordinate system. No two lines in Figure 3 cross one another. Each point in this region of the coordinate system belongs uniquely to a composite of two peaks of a particular height ratio at a particular level of resolution. As the resolution decreases, the characterization of each point in this region becomes more difficult. In practice one might wish to expand the scale of the excess coordinate in order to improve the characterization.

As was seen, the skew–excess lines of the system where the leading peak is smaller lie above the equal peaks line. When comparing Figures 3 and 2, it can be seen that some of these lines will coincide, or cross, or lie above the single peak line. In general, it can be said that points to the left of the single peak line belong definitely to a composite in which the leading peak is the smaller one. Points to the right of the single peak line may belong to a system where the leading peak is either the largest or the smallest. Points falling on the single peak line may, unfortunately belong to double peaks. This is particularly true in the cases where the excess is larger than about 0.5 and the skew is greater than about 0.3.

If, on the other hand, the τ value of the system is known, then by using family of curves such as those shown in Figure 3, one can obtain all the information that is needed to specify the chromatogram. The same type of analysis can be made in the case where the second peak is slightly wider than the first one.

SLOPE ANALYSIS

Slope analysis is, perhaps, easier then moment analysis since the differentiation process is simpler. In slope analysis use is being made of the extrema ratios of the second derivative of the chromatographic peak (*2*). In the practice of chromatography the ratio τ/σ is finite and the second derivative of a single exponentially modified Gaussian peak has three extrema (2 maxima and one minimum). Analytically, the second derivative is best obtained by differentiating twice Equation 2 with respect to t.

$$f''(t) = A \exp\left[\frac{\sigma^2}{2\tau^2} - \frac{(t - t_R)}{\tau}\right]\left\{\frac{\sigma\sqrt{2}}{\tau^3}\int_{-\infty}^{Z} \exp[-x^2]dx - \left(\frac{1}{\tau} + \frac{(t - t_R)}{\sigma^2\tau}\right)\exp\left[-\frac{1}{2}\left(\frac{(t - t_R)}{\sigma} - \frac{\sigma}{\tau}\right)^2\right]\right\} \quad (19)$$

In differentiating Equation 2, use was made of the following relation

$$\frac{d\left(\int_{-\infty}^{f(t)} e^{-x^2}dx\right)}{dt} = \exp\,[-f(t)^2]\,\frac{df(t)}{dt}$$

We can define now three second derivatives extrema ratios. R_1 is the absolute value of the ratio of the 2nd derivative maximum, on the negative side of the actual peak maximum, to the minimum. R_2 is defined as the absolute value of the ratio between the 2nd derivative maximum, on the positive side of the actual peak maximum, to the minimum. R_3 is simply R_1/R_2. As was shown by Grushka *et al.* (*2*), it is most advantageous to plot the data in a R_1 *vs.* R_2 coordinate system. In Figure 4 the solid line describes the behavior of R_1 and R_2 as τ/σ changes. The line starts at the point where $R_1 = R_2 = 0.446$ which corresponds to a true Gaussian. As τ/σ increases, R_1 increases while R_2 decreases. This is to be expected since as the τ/σ ratio increases, the front end of the peak sharpens while the back end flattens. As with the moments analysis, this single peak line is extended only to the point where $\tau/\sigma = 2.6$.

The ratios R_1, R_2, and R_3 are functions of τ/σ only and not of the individual values of the time constant and the standard deviation. Since Equation 19 is a transcendental expression, these three ratios were obtained by computer simulation of the peaks. The integral in Equation 19 is essentially an error function with the argument of $[(t - t_R)/\sigma\sqrt{2} - \sigma/\tau\sqrt{2}]$. McWilliam and Bolton (*9*) in discussing this error function indicated that an approximation must be made when the value of the argument is less then -3. Their approximation, which is based on an asymptotic expression, is sufficiently accurate for reproducing the actual shape of the peak. However, when the ratios in the second derivatives are needed, their approximation is not sufficient since in taking these ratios the errors in the accuracy are much too large. The integral in Equation 19 can be approximated by a six-term asymptotic series, which is an extension of the series used by McWilliam and Bolton (*9*):

$$\int_{-\infty}^{-|x|} \exp(-t^2)dt = \frac{e^{-x^2}}{2|x|}\left[1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \frac{15}{8x^6} + \frac{105}{16x^8} - \frac{945}{32x^{10}}\right] \quad (20)$$

This approximation was used when the argument of the error function was less the -2.4. Above this value, an error function routine, resident in the computer, was used.

The second derivative of a composite peak can be easily found analytically.

$$f''(t) = A \exp\left[\frac{\sigma_1^2}{2\tau^2} - \frac{t - t_R}{\tau}\right]\left\{\frac{\sigma_1\sqrt{2}}{\tau^3}\int_{-\infty}^{Z_1} \exp[-x^2]dx - \left(\frac{1}{\tau^2} + \frac{t - t_R}{\sigma_1^2\tau}\right)\exp\left[-\frac{1}{2}\left(\frac{t - t_R}{\sigma_1} - \frac{\sigma_1}{\tau}\right)^2\right]\right\} + B \exp\left[\frac{\sigma_2^2}{2\tau^2} - \frac{t - t_R - d}{\tau}\right]\left\{\frac{\sigma_2\sqrt{2}}{\tau^3}\int_{-\infty}^{Z_2} \exp[-x^2]dx - \left(\frac{1}{\tau^2} + \frac{t - t_R - d}{\sigma_2^2\tau}\right)\exp\left[-\frac{1}{2}\left(\frac{t - t_R - d}{\sigma_2} - \frac{\sigma_2}{\tau}\right)^2\right]\right\} \quad (21)$$

where

$$Z_1 = \left[\frac{t - t_R}{\sigma_1} - \frac{\sigma_1}{\tau}\right]\frac{1}{\sqrt{2}}$$

and

$$Z_2 = \left[\frac{t - t_R - d}{\sigma_2} - \frac{\sigma_2}{\tau}\right]\frac{1}{\sqrt{2}}$$

As long as d is small enough (*i.e.*, strongly overlapped peaks), this 2nd derivative still has only two maxima and one minimum, and one can utilize the ratios R_1, R_2, R_3. These depend on B/A, on τ/σ_1, τ/σ_2, and on d. The dashed lines in Figure 4 show the behavior of R_1 and R_2 for few systems (few τ/σ values), each consisting of two identical peaks. Each dashed line begins at $d = 0$ where the two peaks completely overlap, and extend to the point where $d = 1.40\ \sigma$ units. At $d = 0$, since the two peaks are identical, the composite is equivalent to a single peak. Thus, all the dashed lines begin at the solid "single peak" line. If the two peaks in the composite are identical, then, as Figure 4 shows, the recognition of double peaks is easy. In practice, of course, the height ratio of the peaks in the composite is not necessarily unity. To investigate the effect of the relative peak height, we centered our attention on the same system discussed in the moments analysis section; namely, $\tau/\sigma_1 = \tau/\sigma_2 = 1.4$.

Figure 5 shows the behavior of R_1 and R_2 of the above mentioned composite system for several peak height ratios. Each

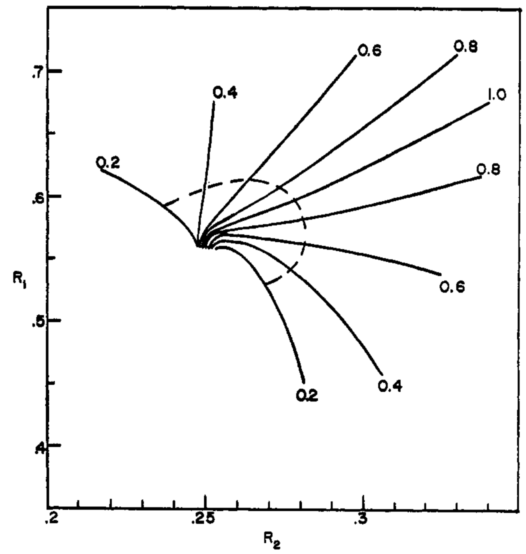

Figure 6. R_1 *vs.* R_2 **of a composite in which** $\sigma_2 = 1.05\ \sigma_1$

The number next to each line indicates the height ratio of the two peaks. The numbers above the line corresponding to $A = B$ **(1.0) are** B/A **values. Numbers below that line are** A/B **values**

solid line extends from $d = 0$ to $d = 1.4\ \sigma$ units. All the lines emanate from a single point, namely a single peak with $\tau/\sigma = 1.4$. It is evident that a region which is uniquely defined exists in this coordinate system. Each point in that region defines the height ratio of the two peaks as well as their separation. To further demonstrate the usefulness of the plot, the dashed line in Figure 5 is a contour of constant d; in this case $d = 1\ \sigma$ unit. Any point on it belongs to a composite peak made of two exponentially modified Gaussians with $\tau/\sigma = 1.4$ at a unique height ratio. As can be seen, when one of the peaks in the composite is very short or at very low resolution levels, the characterization is more difficult. The reason for that was discussed by us previously (*2*).

As is expected, because of the asymmetrical nature of the model, the values of R_1 and R_2 are not symmetrical around the $A/B = 1$ line. In fact, it seems that in the case where $A < B$, and the two heights do not differ by much, the characterization of the double peak is somewhat easier than when $A > B$. Surprisingly, however, it is easier to analyze the system when B/A is between 0.4 and 0.2 than the system where A/B is between these two limits.

As with the moments analysis, a composite made of double peaks of unequal height might cross the so-called single peak line (solid line in Figure 4). Unlike the skew and excess behavior, ambiguity in interpreting the single peak line can be due to either the case where $A > B$ or the case where $A < B$. However, when τ/σ is moderately large, the single peak line crossing occurs mainly when $A < B$.

Figure 6 describes the system were σ_2 is 5% wider than σ_1. This means that the second peak (the more retained one) is slightly more symmetrical than the first one. In general, all the statements and discussions made in connection with Figure 5 hold true here. The noticeable exception is at $d = 0$. It seems that each of the solid lines begins at a different point in the coordinate system. In contrast to the pure Gaussian case (*2*), the points at zero resolution are spread, and, at least in theory, the characterization of the height ratio can be made at zero resolution.

In comparing moments with slope analysis, it can be said that although the changes in the skew and excess are larger than those in R_1 and R_2 (note the scale differences in Figures 3 and 5), the region spanned by the latter coordinates is wider. If the τ value of the system is known, it is perhaps more advantageous to use slope analysis. This is particularly true in routine analysis where standard calibration curves can be easily obtained. If the time constant is not known, then moment analysis might be more desirable, since it can indicate the magnitude of τ. In any event, the exponentially modified Gaussian peak, which is a rather realistic model, lends itself to slope or moment analysis for double peak recognition and characterization.

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Separation and Quantitative Determination of the Yttrium Group Lanthanides by Gas–Liquid Chromatography

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A gas chromatographic method is reported for the separation and subsequent quantitative determination of the yttrium group lanthanides. The lanthanides are synergistically extracted from aqueous solution with the polyfluorinated β-diketone 1,1,1,2,2,6,6,7,7,7-decafluoro-3,5-heptanedione, (H(FHD)), as ligand, and di-*n*-butylsulfoxide (DBSO) as neutral donor. The composition of the extracted species is reported to be R.E.$(FHD)_3 \cdot 2DBSO$. Thermogravimetric analysis of the complexes is reported. Analytical curves were prepared and found usable through a range of 10^{-7} to 10^{-5} gram metal. Individial lanthanides were determined with 97.5% recovery with a relative mean deviation of ±1.5 pph and a relative standard deviation of ±1.9 pph. Mixtures of lanthanides were determined with 97.1% recovery with a relative mean deviation of ±2.3 pph and a relative standard deviation of ±3.1 pph.

CONSIDERABLE INTEREST has been shown in the use of β-diketonates for gas chromatography of rare earths. Chelates of the rare earths with 2,2,6,6-tetramethyl-3,5-heptanedione (*1*) and 1,1,1,2,2,3,3-heptafluoro-7,7-dimethyl-4,6-octanedione (*2*) have been chromatographed. Rare earth chelates have also been prepared with 1,1,1-trifluoro-5,5-dimethyl-2,4-hexanedione (*3*, *4*); however, thermogravimetric analysis indicated decomposition of many of the complexes. In all of these instances, solid anhydrous chelates were prepared and dissolved in a suitable solvent prior to injection into the chromatograph. This type of preparation is unsuitable for quantitative analytical work.

The technique of solvent extraction enables rapid, quantitative formation of metal complexes in the organic phase. Generally the chelates of the rare earths form hydrates which are poorly extracted (*5*). Ferraro and Healy (*6*) found that an organic base such as tri-*n*-butylphosphate would displace the hydrated water and act as a synergist for the extraction of rare earths. This solvent extraction system has been studied in great detail (*7–13*). Butts and Banks (*14*) initially reported the use of these mixed-ligand systems for gas chromatography. Recently Sieck has completed a detailed study demonstrating the chromatographability of several mixed-ligand systems (*15*). Although the chelates of individual rare earths were chromatographed, separation of rare earth mixtures was not achieved and quantitative analysis was not reported.

This paper describes the determination of the yttrium group lanthanides by gas chromatography of the mixed-ligand complexes of 1,1,1,2,2,6,6,7,7,7-decafluoro-3,5-heptanedione, H(FHD). The solvent extraction enables prior separation of the lanthanides from interfering metals (*16*) with subsequent separation and quantitative analysis of the lanthanides.

EXPERIMENTAL

Instrumentation. A Hewlett-Packard Model 5756 B gas chromatograph, equipped with a flame ionization detector and a nickel-63 electron capture detector, was used. A Hewlett-Packard Model 7128 A strip chart recorder was used for developmental work while a 1-mV Bristol recorder equipped with a Disc Integrator, Model 202, was used for the quantitative studies. Helium was used as the carrier gas, and stainless steel tubing was used for all columns and the injection port liner. The following instrument operating conditions were used in the analyses: column, Stainless steel, 20 inch × 0.025 inch o.d., packed with 9.45% Dexsil 300 G.C. (Analabs) on Chromosorb W(100/120 mesh); column temperature, initial 140 to 175 °C and programmed at 6 °C/

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